

A REVIEW OF THE OPERATION OF THE DOUNREAY FAST REACTOR

K. J. HENRY* and A. G. EDWARDS*

1. Introduction

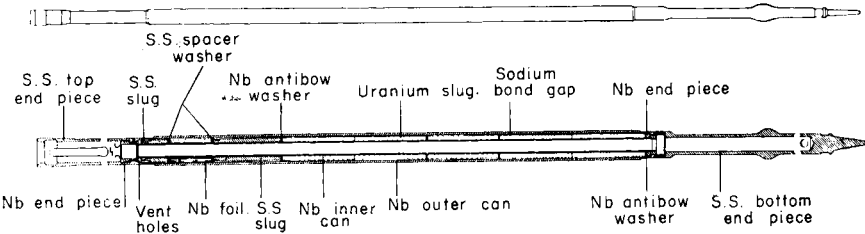
The Dounreay Fast Reactor (DFR) has now operated for several years as a fast flux irradiation facility with a load factor exceeding 60% and by-product electricity generation of 15 MW. The performance of the metal fuel up to 3.4% burnup is excellent. Reactor improvements have included changes to the inner breeder to increase reactivity and enhance local fissile material ratings. Unexpected uranium swelling in the outer breeder has led to modifications there also. Control rod arrangements have been improved, in the light of operating experience, as also have techniques for stability monitoring. Some new instrumentation is described.

There has been a steady improvement in coolant purity, and niobium clad corrosion is no longer a problem. The uranium fuel is exposed to the coolant in DFR, and fission products escape to the blanket gas. There are disadvantages of this arrangement, but they do not outweigh the advantages of the good thermal bond between fuel and can.

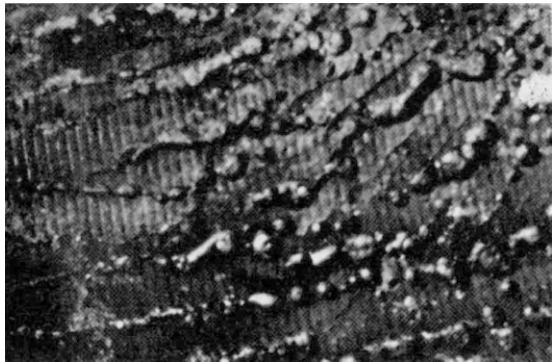
An accurate knowledge of reactor parameters is essential for uninstrumented rigs. The measurement and calculation of neutron flux, gamma flux, and temperature are described. Some reference is made to the three basic types of irradiation experiment, and why they are appropriate to DFR. Finally, possible future improvements and use of the facility are briefly considered.

The Dounreay Fast Reactor first achieved its design power of 60 MW(th) in July 1963 and has since demonstrated its stability and reliability by a series of power runs. The total output of energy from the reactor by the end of February 1966 was 26,900 MWd, and a load factor in excess of 60% was reached during the 12 months prior to August 1965. Electric power was first generated in October 1962, and the 15 MW(e) turbo-alternator has now produced more than 130×10^6 kWh, of which 90×10^6 kWh were exported to the national grid network.

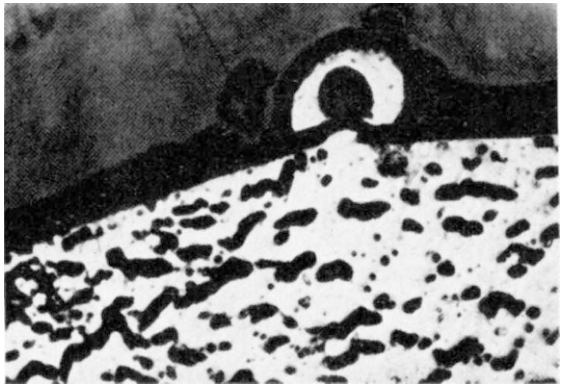
* Dounreay Experimental Reactor Establishment, Caithness, Scotland.



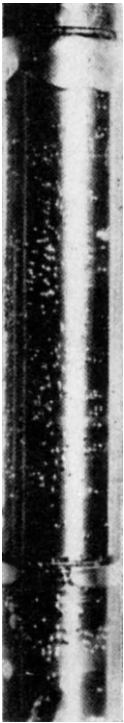
Details of Mk. 3 element.



Globules on fuel surface, burnup 0.75 % heavy atoms.



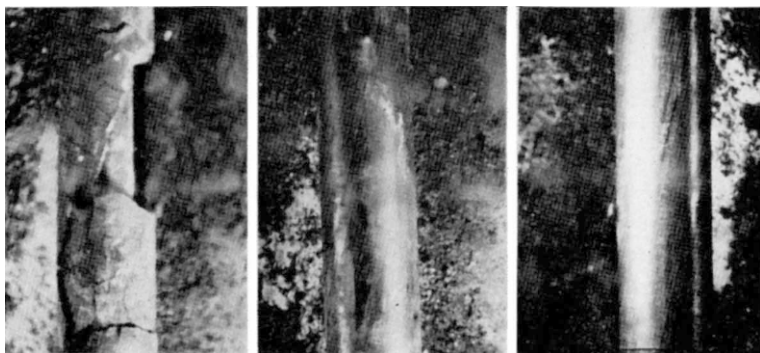
Section through globule on fuel surface showing hollow centre and underlying porosity, burnup 0.9% heavy atoms.



Gas retention in fuel/can annulus of a model Mk. 3 element.

FIG. 1.

FIG. 1 (continued)



Photographs of irradiated fuel slugs from a 3.4% burnup element.

The present function of DFR is to act as a fast flux materials test reactor (MTR) and it is now firmly established in this role. About 100 rigs are loaded into the reactor for each run, designed to explore the characteristics of fuel, clad or moderator in support of either the fast or the thermal U.K. reactor programmes.

2. Performance of the Fuel and Reactor Components

By the end of 1961 the majority of the mechanical and chemical engineering problems which had arisen during the commissioning period had been cleared, and power was raised to 11 MW(th). The remaining uncertainty before full power operation could be achieved was the behaviour of the fuel. Early irradiation results from DFR and from specimens irradiated in a thermal MTR indicated that the fuel and clad operating temperatures of the Mk. 2 elements would need to be reduced. The fuel enrichment of the redesigned Mk. 3 elements was increased to 75% and the fuel thickness correspondingly reduced.

Between the outer can and the fuel there is now a radial clearance of 0.028 in. (0.071 cm) which permits a diametral swelling of 8%, a figure originally believed to correspond to 1% burnup. Elements within the core inner zone have a fuel thickness of 0.1 in. (0.254 cm) and in the outer zone 0.13 in. (0.33 cm), in each case the outside fuel diameter is 0.65 in. (1.65 cm). Details of the element are shown in Fig. 1. To restrict fuel slug movement within the cans, a small clearance only 0.004 in. (0.01 cm) is allowed on the central niobium tube which is itself spaced centrally in the outer tube by four niobium washers. At 60 MW(th) reactor power the design peak fuel temperature is 650°C and the design maximum can temperature is 500°C. The successful performance of this Mk. 3 element to burnup levels of about 2% heavy atoms has been the major factor in achieving high load factors.

During 1962 the reactor was worked up in power to 30 MW(th), and in

May 1963 the first charge of Mk. 3 elements was loaded. The first full power run began in June, and starting at 44 MW(th) the power was increased to 60 MW(th) in steps, continuing for 40 days when the reactor was shut down as planned. Investigation of the kinetic behaviour using the oscillator technique gave results very much as predicted by calculation, the prompt negative feedback coefficient being -4 milliniles/MW. At this stage the primary objective of demonstrating that a power producing fast reactor could be operated safely and reliably had been achieved. Two additional comparatively short runs were completed during 1963; in 1964 three long continuous full power runs were made and two more were completed during 1965, the last of these ending in August after 86 days' operation. The reactor was then shut down to remove a large number of outer breeder elements, and commenced full power operation again in February 1966. No core elements have so far failed in service, and the reactor has never been tripped due to a power transient or temperature excursion.

The Mk. 3 elements were examined at progressive burnup increments. Gross longitudinal and transverse cracking was observed beyond 0.3% burnup, accompanied by crazing of the surface oxide. A second phenomenon was the occasional appearance on the surface of the uranium of hollow globules about 0.02 in. (0.5 cm) in diameter (see Fig. 1) usually near the top end of the fuel beneath a niobium washer. It is believed that gas blanketing, possibly by fission products, resulted in high enough temperatures to induce the extrusion of molten fuel which solidified once it had bridged the gas gap and established good thermal contact with the can. Laboratory water tests suggested that less gas would be trapped if triangular antibowing washers and a higher bond flow were used, and since elements have been thus modified, no further case of the phenomenon has been observed.

More recently, many fuel elements have been taken to 1.8% burnup at peak temperatures of 675°C, in a clad which has reached 530°C. Several pilot elements have reached burnups in the range 2.5–2.9% and three elements attained 3.4%. All these elements were in good condition on examination, with no significant dimensional changes occurring to the inner and outer cans.

Attention has naturally tended to concentrate upon the satisfactory escalation of reactor power, and more recently upon the emergence of the reactor as a unique irradiation facility. Almost taken for granted now is the satisfactory performance of the major components, some as designed and others somewhat modified as a result of accumulating experience.

The liquid metal circuits, with nearly 8000 pipe welds, have not given rise to any concern as to integrity, all leak jackets are continuously monitored and there is confidence not only in the soundness of the circuits but also in the ability to detect any onset of failure. The circuits have now contained the hot liquid metal for 7 years, for the major part of which the coolant has been circulated by the electromagnetic pumps. Apart from very short periods

offline to rectify minor faults in control circuitry or to renew terminal block connexions, all pumps have been operating continuously since mid-1959. This is equivalent to over 2 million pump operating hours without serious failure. During the period of coolant cleanup by dumping at 290°C the maximum permitted winding insulation temperature of 230°C was deliberately exceeded and preparations were made for a programme of rewinding the primary pump stators. This programme was initiated at the first insulation failure which was in mid-1964 and has been completed without any further failures.

Following the installation of a separately pumped cold trap circuit and improved cold traps in three of the primary circuit by-pass loops, there has been no difficulty whatsoever in maintaining a very low oxygen content in the coolant. The figure attained is 6–8 ppm at which level no corrosion is produced in the niobium cans of the fuel elements operating for 90 days at temperatures up to 500°C.

The state of purity of the coolant gave rise to control unit difficulties in the early commissioning stages. The expected improvement in operation was realized when the coolant impurity level was reduced. There have been only minor troubles with the mechanisms since then. A decided improvement in performance of the control system was achieved by modifications which simplified the primary drive units. Every rod has always dropped when required to do so upon the comparatively few occasions of reactor trips from the operating condition, and also when trips are deliberately induced at zero power as part of the rigorous and comprehensive trip test procedure prior to each start-up.

There were no particularly novel features about the instrumentation installation. The guard lines used conventional relay logic and were comprehensively tested by the procedure mentioned above. The safety circuits have never been called upon to protect the reactor.

Major changes have been made to the fuel handling procedure and resulted in a 50% increase in refuelling rate. The refuelling machines have operated very satisfactorily. Operating procedures have been evolved which limit access of the liquid metal coolant to the snout, the cogs and chains being kept clear of the coolant. Up to 200 elements can be handled before it becomes necessary to clean the snout jaws and this is done in an inert atmosphere glove box without allowing access of the atmosphere which would almost certainly result in the need for a complete strip down.

3. Recent Reactor Modifications

Inner Breeder

DFR originally had a substantial radial uranium breeder, and also an upper axial uranium breeder. These were included in the original design when

the reactor was conceived as an experimental power plant and it was considered important that it should have a good breeding ratio. However, breeding ratio is relatively unimportant in an irradiation facility, and the original breeder has now been modified in two ways. The axial uranium breeder has been replaced by one of stainless steel, and the inner three rows of radial breeder elements have been replaced by nickel. In fact the first row consists largely of experiments.

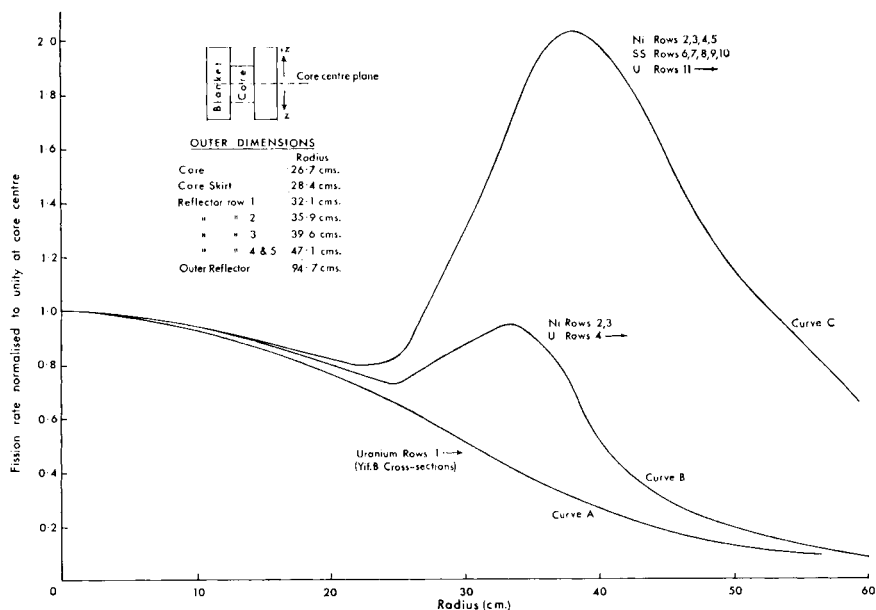


FIG. 2. U^{235} fission rate: calculated radial scans (FD.2 cross-sections).

Besides reducing the breeding ratio, these changes have had the effect of flattening the power profile within the core, softening the spectrum in the breeder to produce a more reactive system and yield increased fissile material ratings in the breeder. It appears that further advantage may accrue from the replacement of more breeder elements by nickel, and in Fig. 2 calculated radial reaction rate scans are shown for the original arrangement (curve A), the present one (curve B) and a possible future one incorporating more nickel (curve C). The calculation of nickel rich systems presents some difficulties, however, and there is some experimental evidence suggesting that predicted ratings are too high.

Outer Breeder

In early 1965, when attempting to remove elements from the outer breeder, it was discovered that swelling of both the natural uranium fuel and distortion

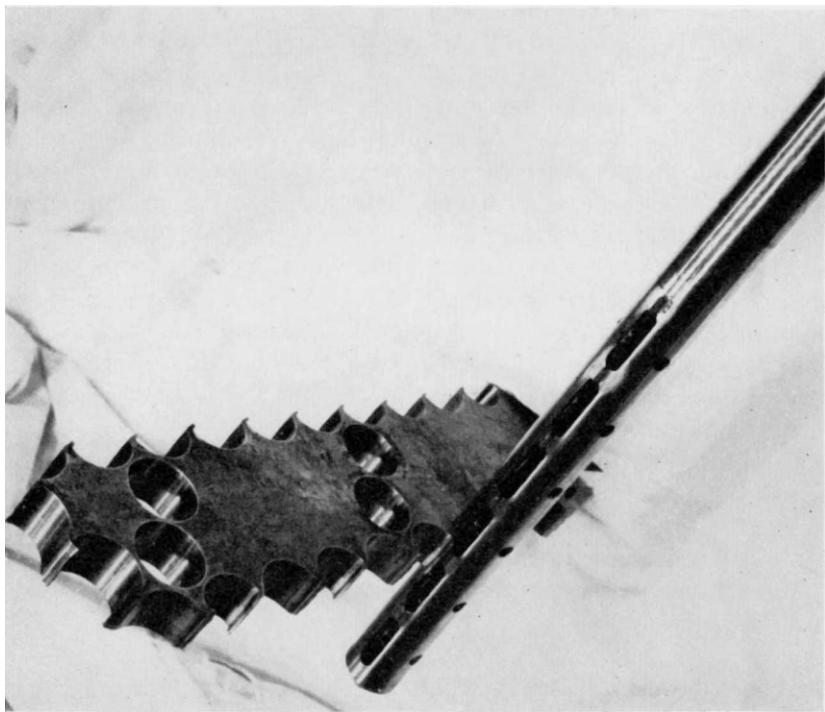


FIG. 3a. Roller cutting tool and specimen of reflector top plate.

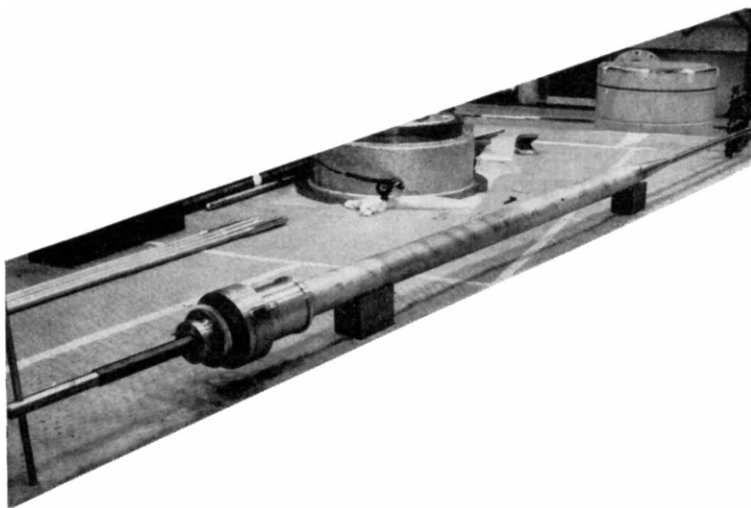


FIG. 3b. Roller cutting tool—general view.

of the stainless steel cladding had occurred although burnup levels were only 0.04% to 0.01% compared with 0.1% previously considered to be a conservative limit.

A large scale attempt at discharging these elements using special equipment was started at the end of September 1965. About 300 elements were removed in this manner but a further 200 or so were found to be swollen by amounts greater than 0.028 in. (0.071 cm) and would not, therefore, pass through the upper tube plates. The decision was taken to cut out small sections approximately 40×10 cm of 2.5 cm thick tube plate in order to remove the distorted elements. Plate cutting equipment (Fig. 3) was designed and five sections were successfully removed. The subsequent removal of the remaining distorted elements was completed in December 1965. Metallographic examination of some of these elements suggests operating temperatures above the γ phase change temperature (760°C) and clad temperatures of 700–730°C. At these temperatures little restraint is offered to uranium swelling resulting from thermal cycling through the phase change, or from the pressure of gaseous fission products even at these low burnup levels.

It is now believed that these temperatures, which are higher than the design figures, occurred because of flow instability in the outer breeder caused by local buoyancy effects.

Alterations to Control Rods

DFR has twelve control rods containing fuel elements. The rate of movement of these rods was fixed on the basis that all twelve rods could, under accident conditions, move together even though the control system was designed to prevent more than six rods adding reactivity simultaneously. The maximum rate of reactivity addition per rod was $2 \times 10^{-4}\%$ per sec and as a spurious trip could result in as much as 10% negative reactivity, recovery of operation following a trip was extremely time consuming.

The control rod drives have now been changed to provide an extremely simple single speed output which permits reactivity addition at ten times the original rate and the rod selector switch has been modified so as to prevent the movement of more than one rod at a time. The overall result of these modifications has been to reduce the time required for start-up by a factor of three, and has facilitated rod shuffling without in any way compromising safety.

Monitoring Kinetic Stability with Reactivity Ramps and Changes of Coolant Flow

The kinetic stability of the reactor is chiefly influenced by its -4 milliniles/MW prompt negative feedback coefficient, arising largely from the axial expansion of the metallic fuel. As a means of confirming that this is

uninfluenced by burnup, use of the reactivity oscillator has now been replaced by impressing reactivity ramps and inferring the negative reactivity from the power change. A small special purpose analogue computer is used for the interpretation. This method has the advantages of speed and the use of a large enough power disturbance to excite fully the non-linear feedback.⁽¹⁾ The power behaviour following the tripping of one pair of coolant pumps can also be analysed by the analogue computer to yield a breakdown of the feedback into flow sensitive and flow insensitive components.

Electrical Supplies

Originally DFR had twelve 150 kVA diesel alternators, each driving a group of two primary e.m. pumps and their associated secondary and steam raising equipment. In addition there were six standby alternators, and another six of 275 kVA capacity to furnish non-essential supplies. Recently, economies both in fuel oil and maintenance manpower have been made by driving eight of the circuits directly from the main turbo alternator, backed by the grid supply. If those should both fail the reactor would be automatically tripped, while four of the e.m. pump circuits would continue to operate from the four diesel sets always in operation. In addition the main circulating sea water pumps now have been changed from diesel drive to electric drive.

New Instrumentation

An advanced new instrument being tested in the reactor is that used to detect subassembly blockages by monitoring the noise in the subassembly outlet temperature. Fluctuations at a frequency of 0.16 c/s are rectified and averaged over 300 sec. The output is coupled to a recorder and an alarm circuit, but since the device is not fully proven it is not coupled directly to trips, though subassembly blockage is one of the more serious of the possible rig induced accidents in the reactor.

In addition there has been installed in the reactor since the middle of 1964 equipment which is designed to detect nucleate boiling of the coolant such as might arise following subassembly blockage. In its present form it consists of three waveguides feeding filter units designed for the range 3–5 kc/s and coupled to the reactor alarm system on a two out of three basis. During a shut-down, a special rig was introduced in which coolant boiling was induced by electrical heating. This was detected on all three channels, but at no other time has the equipment given a positive response.

4. Coolant Purity and Fission Product Contamination

Coolant Purity

Attempts to purify the coolant before operating the reactor at high power⁽²⁾ had, by mid-1962, reduced the oxygen saturation temperature to about

170°C, corresponding to 16 ppm. A large external pumped loop containing two cold traps was then installed and this, together with three improved cold trap baskets in the by-pass circuits, quickly reduced the saturation temperature to 140°C. Since this was the lowest temperature attainable in the by-pass circuits, these baskets were hoisted clear of the NaK surface and held in reserve.

When operating, the external cold trap loop maintained the plugging temperature at 120–130°C. With the loop out of circuit the plugging temperature rose slowly to about 150°C, indicating that an oxide source was still present in the reactor. At this time (1962) the niobium fuel can corrosion rates were about 0.010 in. (0.254 mm) per month at 500°C.

Prolonged operation of the loop, probably assisted by the gettering action of the extremely large uranium surface which was exposed to the coolant finally reduced the plugging temperature to less than 100°C by the middle of 1964 with evidence that the oxide source no longer existed. At this time no measurable fuel can corrosion occurred in 3 months at a temperature of 530°C. In 1964 equipment was installed for obtaining NaK samples from the reactor at full power, which were fully characteristic of the coolant under these conditions. Analysis of these for oxygen confirmed the plugging temperature results, figures of 10 ppm being obtained.

The rhometer, which measures the resistivity of the coolant, has been used on the primary coolant circuit throughout the full power operating period and has been useful in giving a trend indication of coolant purity. Comparison of its readings with those of the plugging meter over a fairly long period confirm its suitability for this purpose.

Consequences of Using Vented Fuel

An unusual feature of the DFR fuel design is that the uranium fuel alloy is in direct contact with the coolant. This design ensures good thermal contact between the clad, coolant, and fuel but as a result permits the escape of fission product gases via the coolant to the blanket gas. The external gas blanket pipes had not originally been shielded because the estimates of activity were so subjective. By the end of 1962 experience showed that shielding was necessary.

Air borne activity from small leaks also proved troublesome and the remedy for this was to modify as many as possible of the flanged joints in the system by fitting them with double "O" rings. This measure reduced the air borne activity by about a factor of 100. From measurements made of blanket gas activity it was estimated that the leak rate even before the modifications was only of the order of 1 litre per hour at NTP. Air activity in the containment building was mostly rubidium 88 ($T_{\frac{1}{2}} = 17.8$ min) and caesium 138 ($T_{\frac{1}{2}} = 32$ min).

Typical analyses of samples of blanket gas taken under full power operation are:

A^{41}	8×10^{10} dpm/litre
X_{135}	4×10^{10} dpm/litre
X_{133}	2×10^{10} dpm/litre
Total	14×10^{10} (equivalent to about 60 mc/litre).

A coolant gamma spectrum measurement 6 days after removal from the reactor is given in Table 1.

TABLE 1. TYPICAL GAMMA SPECTRUM OF LIQUID METAL COOLANT 6 DAYS AFTER SAMPLING AT FULL POWER

Energy MeV	Possible isotopes	Activity dpm/g
0.14	Ce 143/144	1.5×10^6
0.22	Te ¹³²	2.7×10^6
0.36	I ¹³¹	1.52×10^7
0.50	Ru ^{103/106}	1.5×10^6
0.67	I ¹³² /Cs ¹³⁷	1.63×10^7
0.76	Zr/Nb ⁹⁵	5.18×10^6
1.6	Ba/La ¹⁴⁰	1.95×10^7

At full power the short lived fission product activity is principally from caesium 138 and is about twice that of the sodium 24. Analyses of the coolant for plutonium and uranium indicate that these, if present, are below the levels of detection of 3×10^{-2} ppm and 10^{-4} ppm respectively.

The activities found in the coolant and blanket arise mainly because of the design of the fuel element. While causing some slight inconvenience it in no way detracts from the advantages of using vented fuel.

5. Reactor Parameters

In order to calculate rig conditions, neutron and gamma flux distributions in the reactor must be known accurately. The present situation is described below.

Neutron Flux

The main calculation tool is the CRAM diffusion theory code used in RZ geometry. A full length model involves thirty different regions represented in a 36×26 mesh. The flux in the nickel-rich inner breeder regions is quite soft,

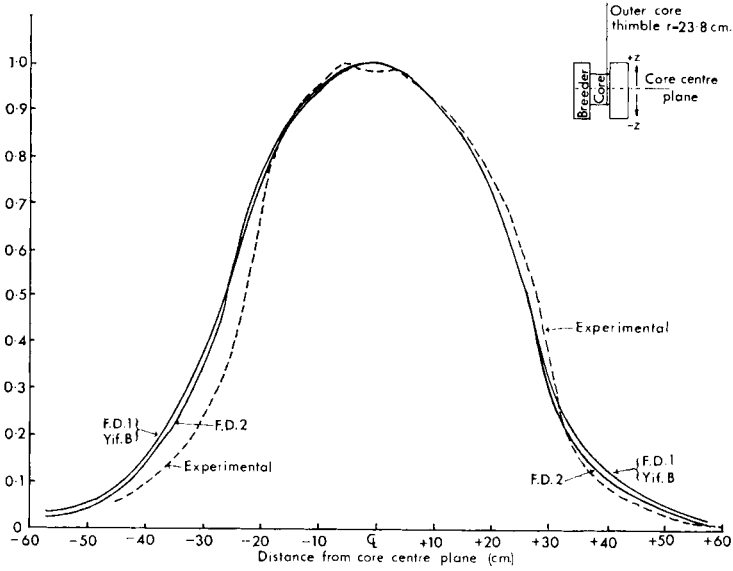


FIG. 4a. U^{238} fission rate vertical scan in outer core thimble compared with calculations using different cross-section sets.

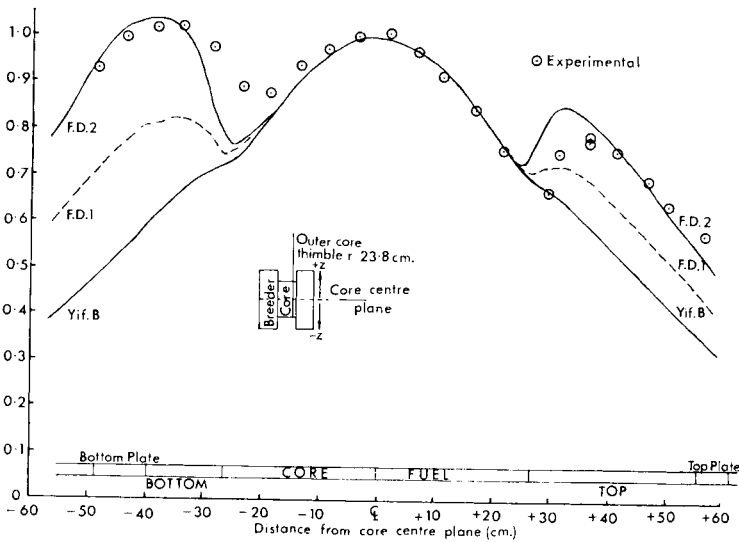


FIG. 4b. U^{235} fission rate vertical scan in outer core thimble compared with calculations using different cross-section sets.

and the cross-sections used cover the energy range down to 0.4 eV in a 15 group condensation of the 33 group FD.2 set. Such a calculation takes several hours of IBM 7090 time, and is near the effective maximum size which the computer can handle, though for many purposes a much simpler model is adequate.

In Fig. 4 the calculated axial reaction rate scan for U^{238} and U^{235} made with FD.2, FD.1 and YIF.B cross-section sets are shown. The FD.1 cross-section set differs from YIF.B in having a number of additional groups extending the coverage to 0.4 eV. The FD.2 set is based on re-evaluated data, and in particular incorporates reduced nickel capture in the low energy range. The core edge is one of the few positions available for experiment, but here the inadequacy of a cylindrical calculation model of a hexagonal reactor is very apparent. The limited experimental facilities have been described by Smith.⁽³⁾ In the time allotted for this work scans are made using U_5 , U_8 , Pu_9 and BF_3 detectors in the various thimbles for different control rod configurations. The experimental scan in Fig. 4 revealed the inadequacy of the then current calculation, which was eventually traced to poor nickel data.

Foil irradiations using long half-life materials are attractive as a means of establishing flux profiles provided they can extend over the full length of an 80 day irradiation period. Samples of reasonable size may become inconveniently active, but the technique has been applied amongst others to the $Co^{59}(n\gamma)$ Co^{60} reaction for low energy flux, and to $Cu^{63}(n\alpha)$ for the high energy flux. Irradiated UO_2 samples have also been analysed for total number of fissions.

Another technique is to measure, by gamma scanning, the distribution of those fission products in fuel elements which are believed not to migrate, and if several sets of data are averaged, reasonable agreement with theory is obtained in the core, though there are anomalies elsewhere. The data, though numerous, are not sufficiently accurate to indicate other than fairly gross (say 5%) errors in calculation.

Gamma Flux

The dominant heating agency for non-fissile materials is gamma absorption which amounts to about 5 W/g for steel at the reactor centre at full power. Gamma heating rates show a more marked fine structure than fission rates and in calculating them reactor geometry must be allowed for in some detail. The method is to calculate the distribution of a pseudo flux by numerical integration of an exponential attenuation kernel from which heating rates are derived directly. The method has been checked by a Monte Carlo calculation, and compared with experiment. This calculation remains less adequate, however, than its nuclear counterpart and offers accuracies

no better than $\pm 50\%$. Where heating of a rig is solely by gamma absorption preliminary calibration experiments may be undertaken.

Temperatures

The pressure drop across the core is about 10 lb/in² (0.7 kg/cm²), and the flow rate in channels can be established by the combined use of calculation and water rig tests. Temperature distributions are then derived using FRET, a computer programme which allows for a linear dependence of thermal conductivity of fuel on temperature. Special elements are dealt with individually. In the breeder, pressure drops are less than 10% of those in the core, and buoyancy forces have so serious an affect on flow that detailed calculations are not attempted.

Experimentally, temperatures up to about 300°C are measured by post irradiation examination of fusible alloys. Above 400°C templuges are similarly used, offering laboratory accuracies of $\pm 10^\circ\text{C}$, though rather poorer results are obtained in the reactor environment.

Setting Up Reactor Power

The objective is to maintain the central flux constant throughout a run, despite an 8% increase in core size as rods are brought in to compensate for burnup. The installed instruments are sensitive to core configuration, so the required power is determined by measuring temperature rises across selected fuel elements and rigs. Partly because of poor siting of existing thermocouples, the error in power determination is estimated to be $\pm 6\%$ but this is being improved. The total power swing during a run is typically 58–62 MW.

Hot Spot Factors for Fuel Element and Rigs

Factors such as engineering tolerances introduce into operation of a rig hot spot factors which are specific to that rig. In addition there are reactor parameter uncertainties. Central temperature for an oxide pin is sensitive to local rating, and clad temperature to the total power generated in that channel, though generally maximum uncertainties in the core are found away from the centre, particularly near displaced control rods.

An estimated hot spot factor for the clad of a typical single fuel pin rig in the core is $\sim 80^\circ\text{C}$, but possible improvements might reduce this to less than 50°C.

6. Experimental Irradiations

As an irradiation facility DFR has been extremely successful for work in support of both the fast and the thermal reactor programmes. In some cases,

in fact, its superiority to thermal flux MTR's is such that the two can hardly be classed together. The reasons for this are considered below for the case of fuel, clad, and graphite irradiations.

Fuel

When test irradiating proposed fast reactor fuel, it is desirable to reproduce the conditions of the conceptual fast power reactor under study. Where complete subassemblies are concerned, a fast flux MTR is essential to avoid the flux depression effects, and hence shielding of the central pins which would be marked in a thermal MTR. Even with individual pins there will be significant depression at the pin centre in a thermal flux, and both temperature profiles and fission product distributions will be affected. It may also be noted that in a metal cooled test reactor it is easy to reproduce clad temperature conditions which are adequately representative of those encountered in the conceptual fast power reactor under study. By contrast, a sodium cooled rig would be required to reach the same ends in a water cooled and moderated MTR.

The DFR flux is less than that in conceptual fast reactors and in order to get the correct ratings in a PuO_2/UO_2 fuel, the uranium is enriched, which is a quite satisfactory expedient unless the small differences in fission product yields of uranium and plutonium are significant. Experimental subassemblies can then be irradiated under fast power reactor conditions, and the successful performance of such subassemblies in DFR has built up confidence in the U.K. design for a 600 MW(th) prototype fast reactor.

It is not always necessary or desirable to use complete experimental subassemblies; a large number of individual pins have been irradiated. An early form of vehicle for this purpose incorporated one pin, and a later design three. Whilst the reactivity penalty per pin in such experiments is higher than for subassembly irradiation, they have been found of great value in view of their relative simplicity to mount and interpret. As with subassembly irradiations, proper design can give correct ratings and representative clad conditions.

The number of such rigs in the reactor has steadily increased, and is now well in excess of thirty.

Clad

Where clad is concerned it is more difficult to speak of equivalence between DFR and other reactors because the damage mechanisms are sensitive in different ways to neutron energy. Thus damage from threshold np and n reactions depends upon the MeV flux component, whereas displacement damage in the crystal lattice is most sensitive to neutrons in the hundreds of

KeV energy range, and damage due to n reactions with impurity boron is a function of the B_{10} cross-section. It is found that where the first mechanism is concerned, rates in DFR and a high flux thermal MTR are roughly comparable. DFR damage rates by the second mechanisms are markedly higher, and by the third mechanism lower than in an equivalent thermal MTR. Clad evaluation therefore requires irradiation in both DFR and its thermal counterpart for a proper appreciation of how damage rates will depend on dose and position in the reactor.

Graphite

At first sight it is surprising that DFR has been so successful in irradiating graphite and zircaloy, both materials of interest in the thermal reactor programme. Irradiation damage in these materials is due to the displacement of atoms from their lattice positions by the impact of fast neutrons. In the low energy range the number of displacements per neutron increases with the neutron energy, but at higher energies other types of atom-neutron interaction take place and the likelihood of atomic displacements is reduced. The neutron energy which is predicted to be most effective in causing displacements depends somewhat on the details of the theory used, but is in the hundreds of KeV range. That is, it is comparable with the energy at which the DFR flux peaks. From this theory can be calculated energy dependent damage cross-sections, and from them damage rates, by weighting the multi-group cross-sections with the multi-group flux in the usual way.

Damage rates have been assessed in this way for graphite exposed in AGR, in the high flux thermal test reactor DMTR, and in DFR and they are found to stand in the ratio 1:2:40. Less than a year's irradiation in DFR thus corresponds to a lifetime's dose in AGR. A similar picture emerges if zircaloy is taken instead of graphite, and in addition the damage rate ratios are found to be insensitive to the theory of atomic displacement employed.

This feature of fast flux irradiation has been exploited by Simmonds in graphite studies, and the results obtained have been of great value in assessing the economics of the AGR system.

Temperature Adjustments

For non-fissile irradiations temperatures are adjusted in the range of interest by one of two techniques. The first depends upon the use of thermally insulating gas gaps which will raise the specimen temperature because of its own gamma heating. In the second technique, the specimen is mounted in a gagged tube downstream of a fissile pin so that it is immersed in coolant at the required temperature. Thermal impedance for fissile pins has been provided by gas gaps or thick walled stainless steel tubes.

7. The Future of the Dounreay Fast Reactor

DFR is now firmly established in its role of a fast flux MTR, and load factors exceeding 60% have regularly been reached. The demands on its irradiation space by the U.K. fast reactor programme is now slackening but it is envisaged that there will be sufficient work from other quarters within the Authority or outside it to keep the facilities fully subscribed.

The majority of experiments so far have used simple uninstrumented rigs since the extraction of electrical leads from the reactor is difficult. This does not represent the disadvantage it would in a thermal MTR, in view of the relative stability and ease of calculation of the flux in a fast system, but nevertheless there is a demand for more instrumented rigs and these are being provided.

Theory indicates and experience confirms that a fast flux MTR has great value as a general purpose MTR. Where DFR is concerned the passage of time will yield further experience, more instrumented rigs, a better knowledge of reactor parameters, and perhaps a higher flux, all of which will enable the project staff to offer both reactor technologists and materials scientists an increasingly powerful tool in solving their problems.

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